CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 6: Solving Problems Based on Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Problems on speed, distance and time

1. A couple drive 350 km to a hill station. Heavy rain reduces their average speed by 20 km/h, and the journey takes 2 hours longer. Find (a) the original speed of the car (b) the time taken at the reduced speed. ICSE 2025, 4 marks

SOLUTION

(a) Let the original speed be x km/h. $\frac{350}{x-20} - \frac{350}{x} = 2$

$350x - 350(x-20) = 2x(x-20)$, so $7000 = 2x^2 - 40x$ and $x^2 - 20x - 3500 = 0$

$(x-70)(x+50) = 0$, so $x = 70$ (speed cannot be negative).

(b) Reduced speed $= 50$ km/h, so the time is $\frac{350}{50} = 7$ hours.

Answer: (a) 70 km/h (b) 7 hours

2. A man travels 100 km at a uniform speed of x km/h. At a speed 5 km/h faster, the journey would have taken 1 hour less. Find x. ICSE 2023, 3 marks

SOLUTION

$\frac{100}{x} - \frac{100}{x+5} = 1$

$100(x+5) - 100x = x(x+5)$, so $x^2 + 5x - 500 = 0$

$(x+25)(x-20) = 0$, so $x = 20$ (speed cannot be negative).

Answer: 20 km/h

2 Problems on numbers

3. Two natural numbers differ by 7 and their product is 450. Find the numbers. ICSE 2020, 3 marks

SOLUTION

Let the smaller number be x. Then $x(x+7) = 450$, so $x^2 + 7x - 450 = 0$.

$(x+25)(x-18) = 0$, so $x = 18$ (a natural number cannot be negative).

Answer: 18 and 25

MathSetu

4. Two consecutive multiples of 3 are natural numbers with a product of 810. Find them. ICSE 2019, 3 marks

SOLUTION

Let the numbers be $3x$ and $3x+3$. Then $3x(3x+3)=810$, so $x(x+1)=90$.

$x^2+x-90=0$, so $(x+10)(x-9)=0$ and $x=9$.

Answer: 27 and 30

3 Problems on cost and on work

5. ₹7500 is divided equally among some children. With 20 fewer children, each would have received ₹100 more. Find the original number of children. ICSE 2018, 3 marks

SOLUTION

Let there be x children. $\frac{7500}{x-20}-\frac{7500}{x}=100$

$7500\times 20=100x(x-20)$, so $x^2-20x-1500=0$

$(x-50)(x+30)=0$, so $x=50$.

Answer: 50 children

4 Problems on ages

6. The ages of two brothers add up to 47 years, and the product of their ages is 550. Find their ages. ICSE 2017, 4 marks

SOLUTION

Let the elder brother be x years old. Then the younger is $(47-x)$ years old.

$x(47-x)=550$, so $x^2-47x+550=0$

$(x-25)(x-22)=0$, so $x=25$ or 22. The elder is 25.

Answer: 25 years and 22 years

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.